

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	31 January 3035
Team ID	LTVIP2025TMID54870
Project Name	DocSpot: Seamless Appointment Booking for Health
Maximum Marks	4 Marks

Technical Architecture

 **Use Case:** *Online Doctor Appointment Booking System with AI Chatbot Integration for Recommendations and Smart Scheduling*

Architecture Diagram:

You can draw this using tools like draw.io, Lucidchart, or Figma.

Suggested flow:

```
csharp
CopyEdit
[User Device]
    ↓
[Web / Mobile UI]
    ↓
[Backend Logic (Python Flask)]
    ↓
[ML Model - Doctor Recommendation]
    ↓
[Database (MySQL/Cloud DB)]
    ↓
[Cloud Deployment - IBM Cloud / AWS]
```

 **Table 1: Components & Technologies**

S · N o	Component	Description	Technology
1	User Interface	Web UI for patients & doctors	HTML, CSS, JavaScript, Bootstrap
2	Application Logic-1	Backend logic for login, register, search, book appointment	Python (Flask)
3	Application Logic-2	Speech-to-text for voice appointment booking	IBM Watson Speech to Text
4	Application Logic-3	Chatbot for patient queries and appointment scheduling	IBM Watson Assistant
5	Database	Store user, doctor, booking data	MySQL
6	Cloud Database	Cloud-based storage for high availability	IBM Cloudant
7	File Storage	Store reports and patient images	IBM Object Storage
8	External API-1	Get live doctor availability and sync calendar	Google Calendar API
9	External API-2	Verify identity using government ID	Aadhar API
10	Machine Learning Model	Recommend doctors based on symptoms & history	Sklearn + Trained Random Forest
11	Infrastructure	Deployment on IBM Cloud / AWS EC2 using Flask server	IBM Cloud Foundry / AWS EC2

 **Table 2: Application Characteristics**

S · N o	Characteristics	Description	Technology
1	Open-Source Frameworks	Python Flask, Bootstrap, jQuery, Sklearn	Flask, Bootstrap, jQuery
2	Security Implementations	SHA-256 Password Encryption, HTTPS, OAuth2, IAM Roles	OpenSSL, OAuth2, JWT, IAM

3	Scalable Architecture	Modular architecture using microservices & containerization	Docker, Kubernetes, Microservice Approach
4	Availability	Load balancers with cloud servers, NoSQL fallback	NGINX, AWS Elastic Load Balancer, MongoDB
5	Performance	Redis caching, CDN for static content, optimized SQL queries	Redis, Cloudflare CDN, Optimized SQL Joins

References

- [IBM Architecture Center](#)
- [How to Draw Architecture Diagrams](#)
- [C4 Model](#)
- [IBM AI Patterns](#)
- [AWS Architecture Best Practices](#)